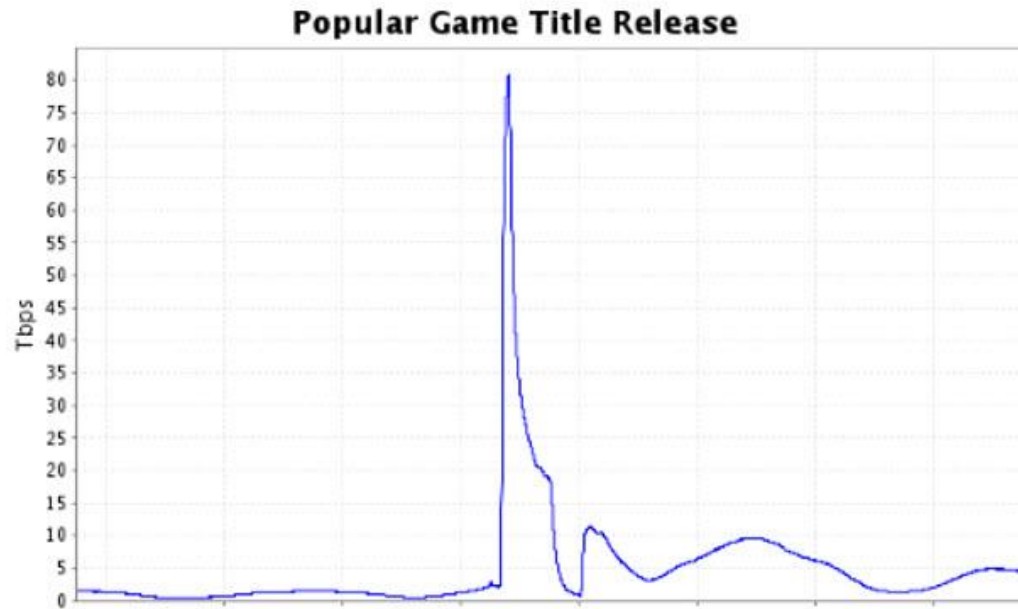


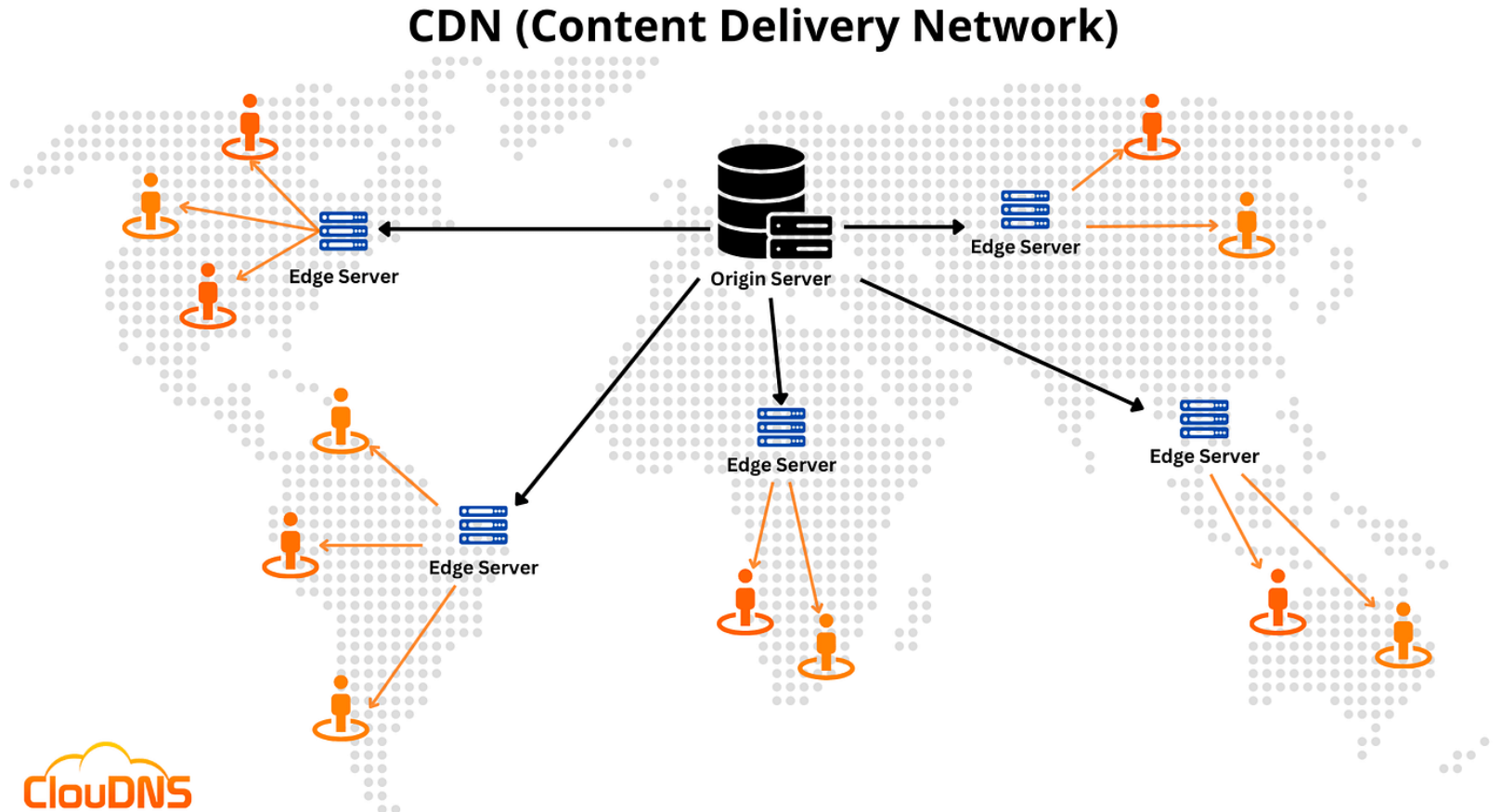
Learned CDN caching

Mar 05 2026

Challenge of large-scale events



Content Delivery Networks



CDN challenges

1. Where to place servers (PoP)?
2. Handling failures
3. Consistency issues
4. Caching different types of objects
5. Routing requests

CDN caching

- Cache eviction + *admission*
- Admission policies??
- Challenge: variable sized objects
 - Cache 1 big object with 10 accesses or cache 5 small objects each with 2 accesses?

Bandit algorithms

- When can bandit algorithms fail?
 - Delayed feedback
 - Noisy rewards
 - Each arm's distribution is varying
 - Lot of arms or difference between arms is small